

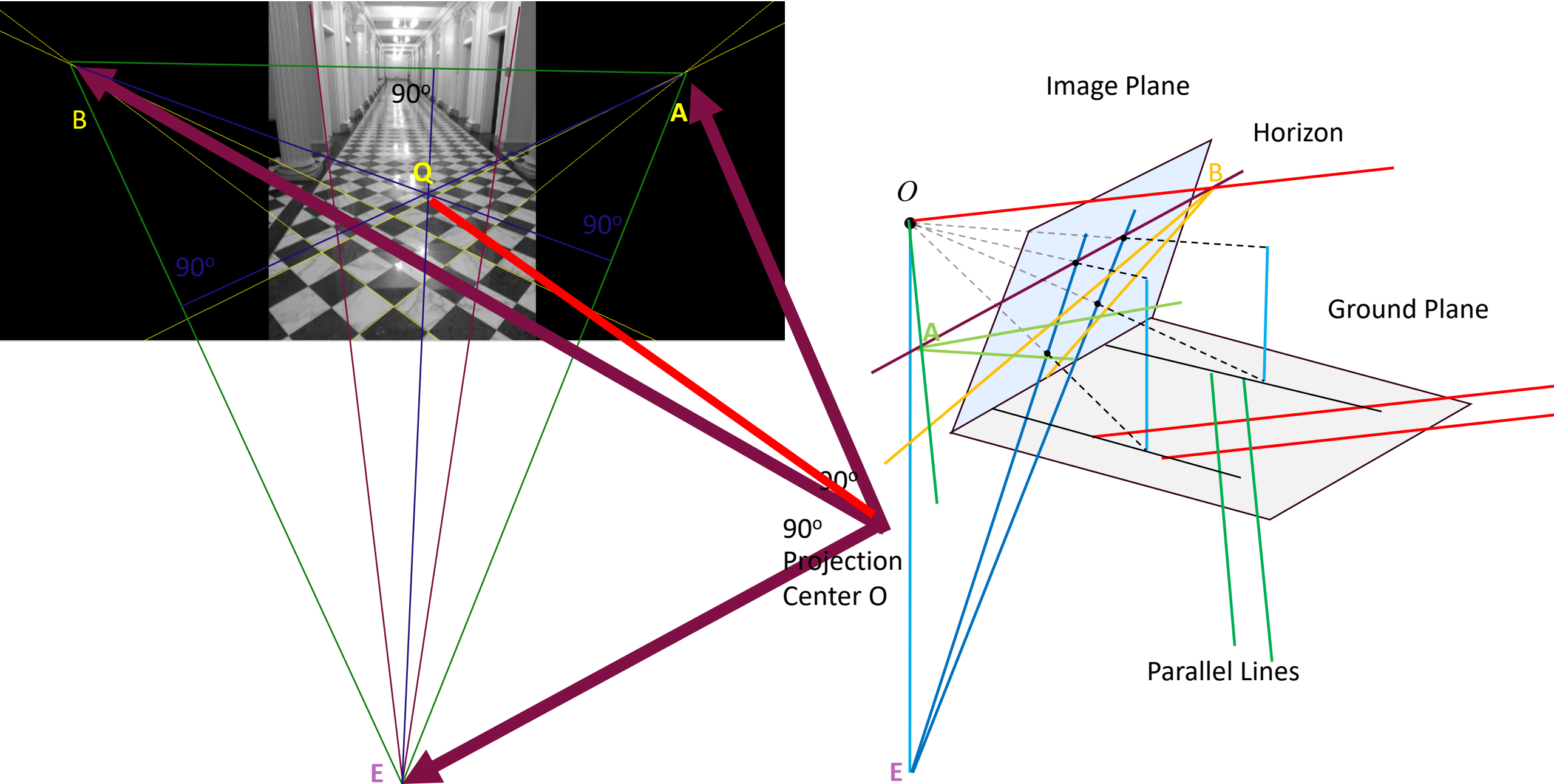
CIS 5800

Machine Perception

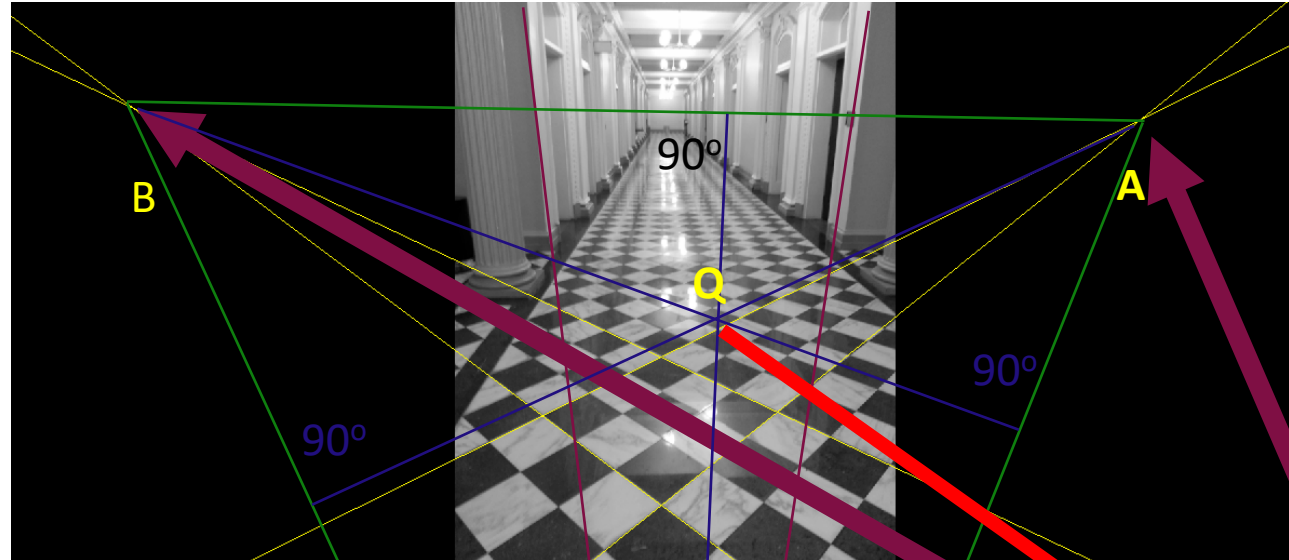
Instructor: Lingjie Liu

Lec 7: Feb 12, 2026

Recap: Computing Intrinsic K from Vanishing Points



Recap: Let Q be the orthocenter of the triangle ABC

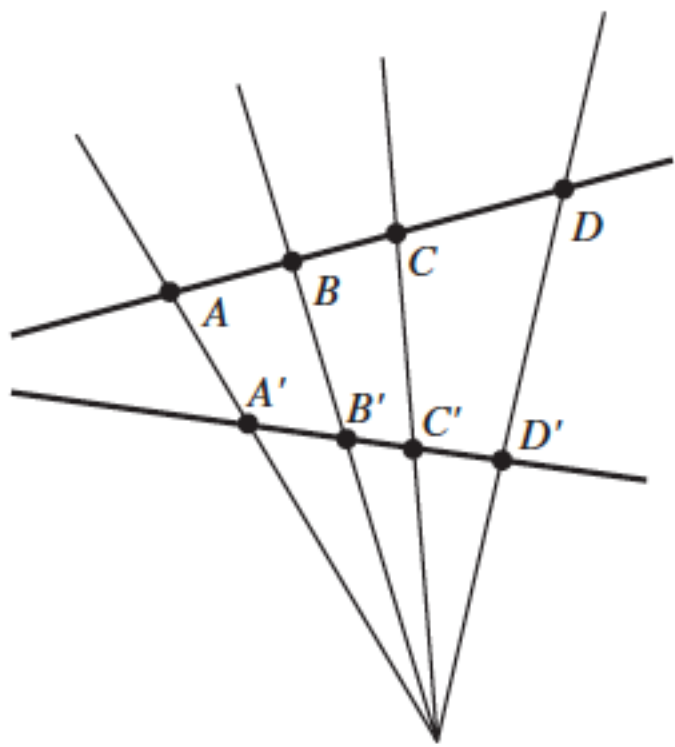


Theorem from Euclidean Geometry:

If Q is the orthocenter of ABE and all three angles AOB , BOE , and EOA are right angles, the OQ is perpendicular to ABE plane!

OQ is the principal axis and ABE is the image plane, hence, Q is the principal point

Recap: Cross Ratios of Collinear Points



Given four points A, B, C, D , we define the cross-ratio of their distances as

$$CR(A, B, C, D) = \frac{AC}{AD} : \frac{BC}{BD}.$$

$CR(A, B, C, D)$ remains invariant under projective transformations

$$\frac{AC}{AD} : \frac{BC}{BD} = \frac{A'C'}{A'D'} : \frac{B'C'}{B'D'}$$

Denote $\mathbf{x}'_i = (x'_i, 1)^T$ and $\mathbf{x}_i = (x_i, 1)^T$. Suppose $\mathbf{x}'_i = H\mathbf{x}_i$. Note $|H| = \det(H)$. Then

$$x'_i - x'_j = |\mathbf{x}'_i \ \mathbf{x}'_j| = |\lambda_i H\mathbf{x}_i \ \lambda_j H\mathbf{x}_j| = \lambda_i \lambda_j |H(\mathbf{x}_i \ \mathbf{x}_j)|$$

$$= \lambda_i \lambda_j |H| |\mathbf{x}_i \ \mathbf{x}_j| = \lambda_i \lambda_j \det(H) (x_i - x_j)$$



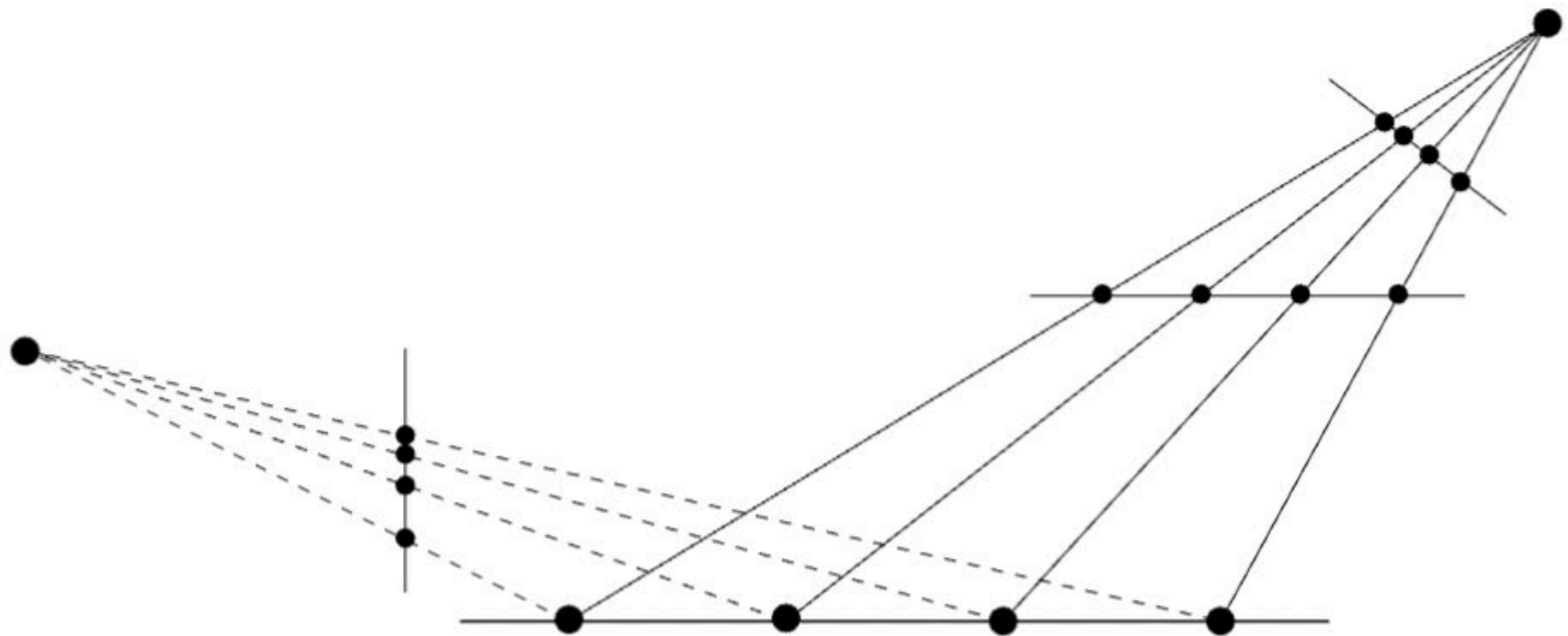
Pappus (290-350 AD)



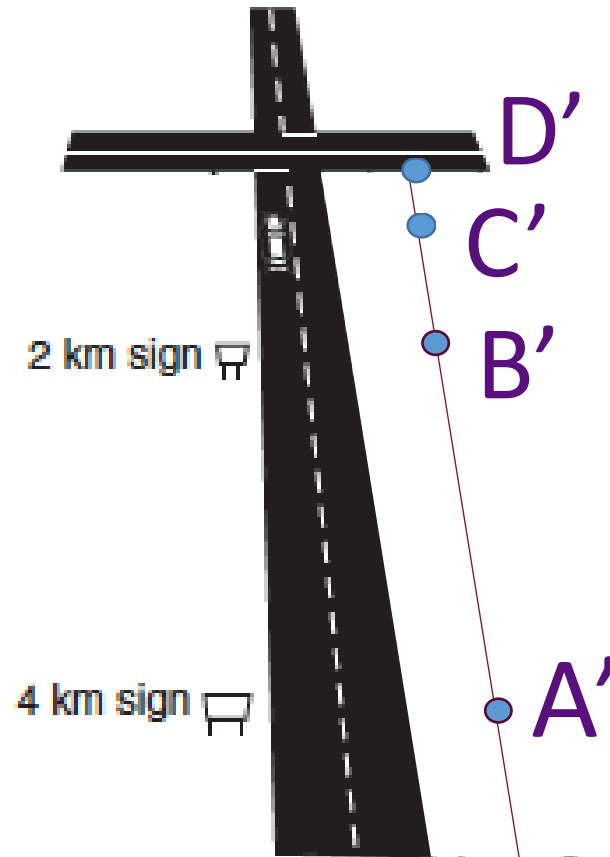
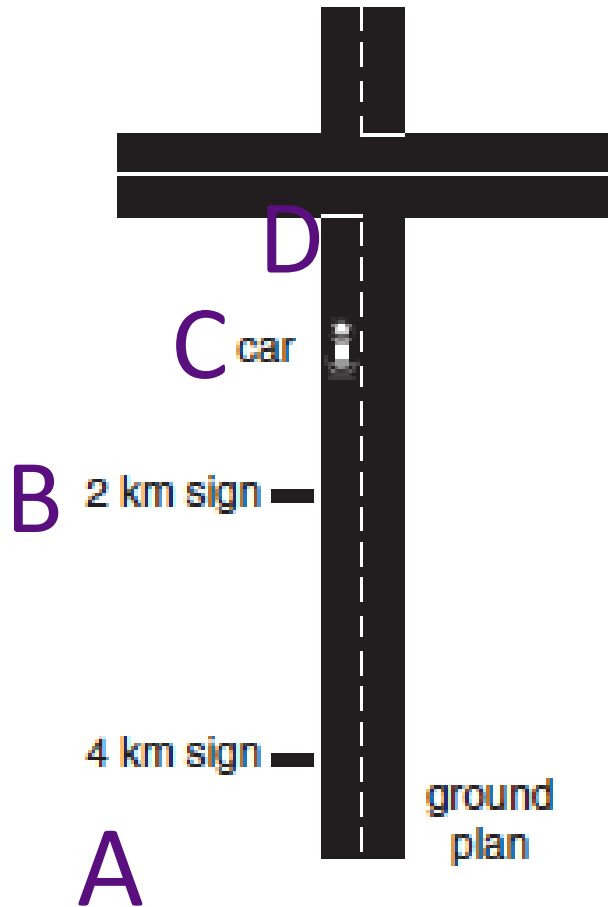
Girard Desargues (1591-1661)

Note: C
but it is
e.g., CR

Same cross ratio for all these 4 point sets!



Example: Cross ratios for metrology



How far is car from intersection:

CD=?

A'D'=300 pixels

A'C'=275pixels

B'C'=50 pixels

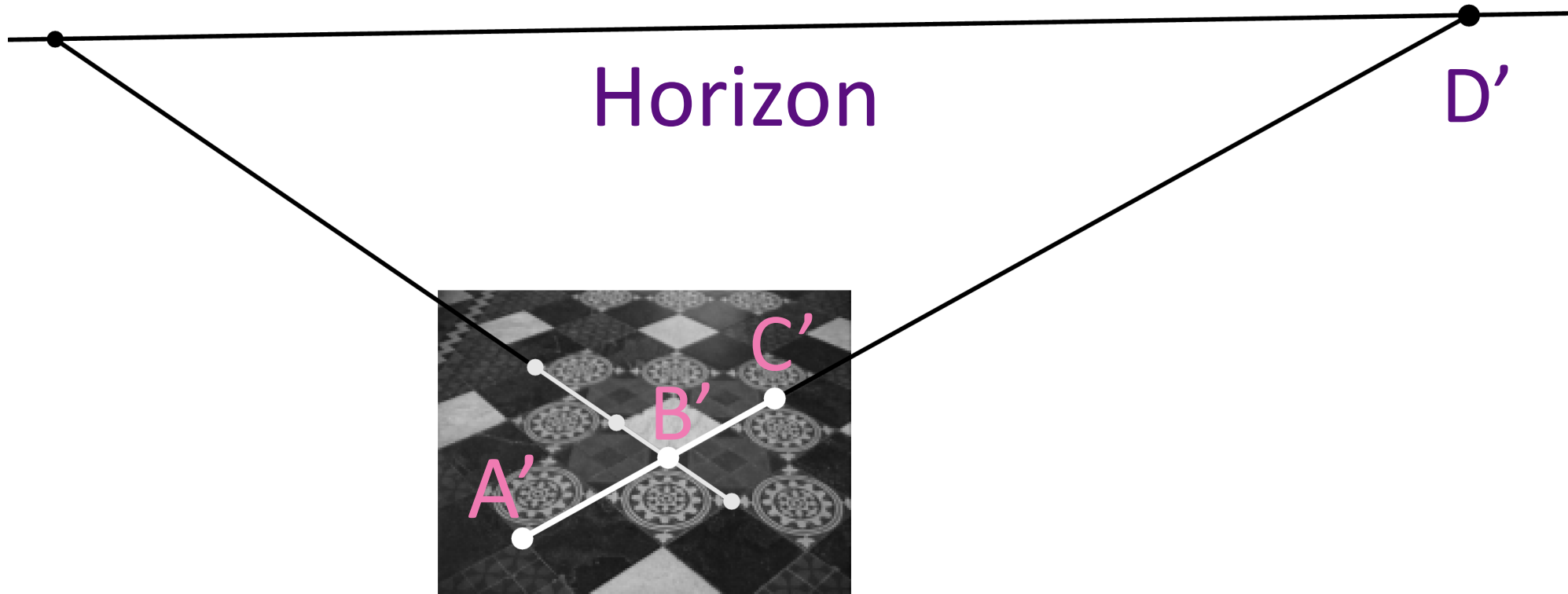
$$\frac{A'C'}{A'D'} : \frac{B'C'}{B'D'} = \frac{275}{300} : \frac{50}{75} = 1.375$$

$$\frac{AC}{AD} : \frac{BC}{BD} = \frac{4-CD}{4} : \frac{2-CD}{2} = 1.375$$

$$CD = 0.857km$$

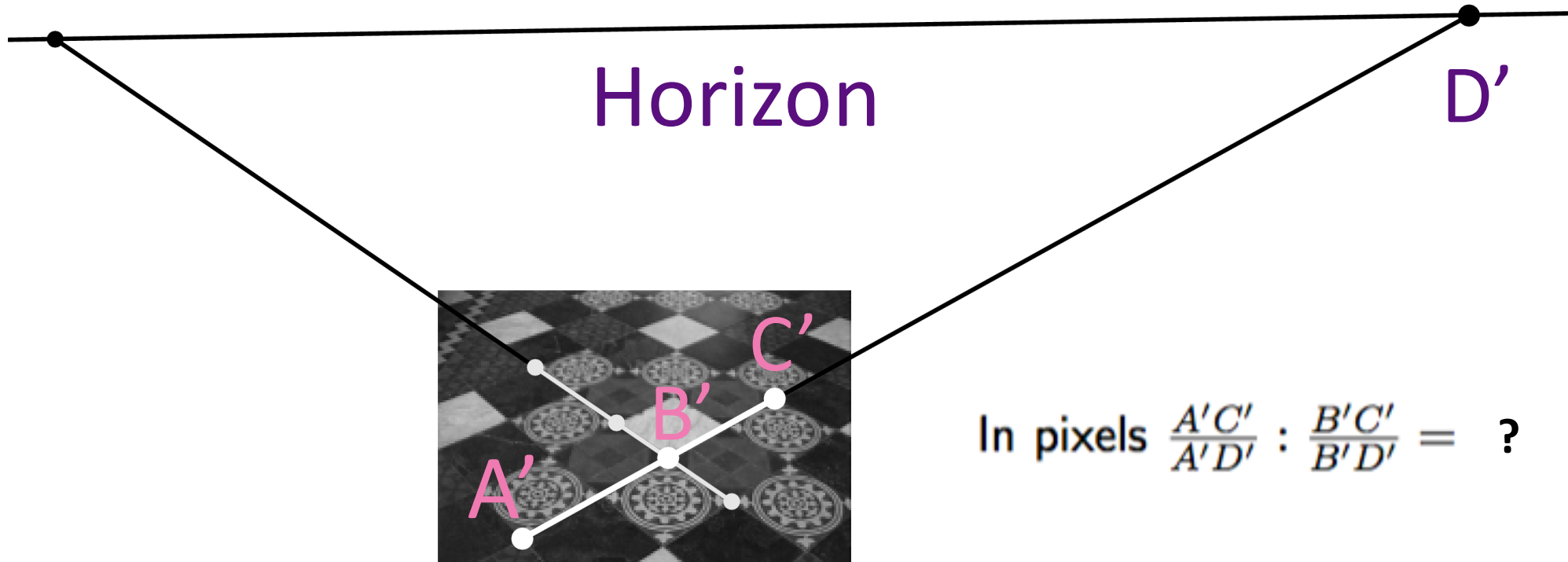
Note: all distances in image are measurable. And 2 distances in world are given.

What happens when one of the points is at infinity?



While D' is a finite point, D on the original plane is at infinity !

What happens when one of the points is at infinity?

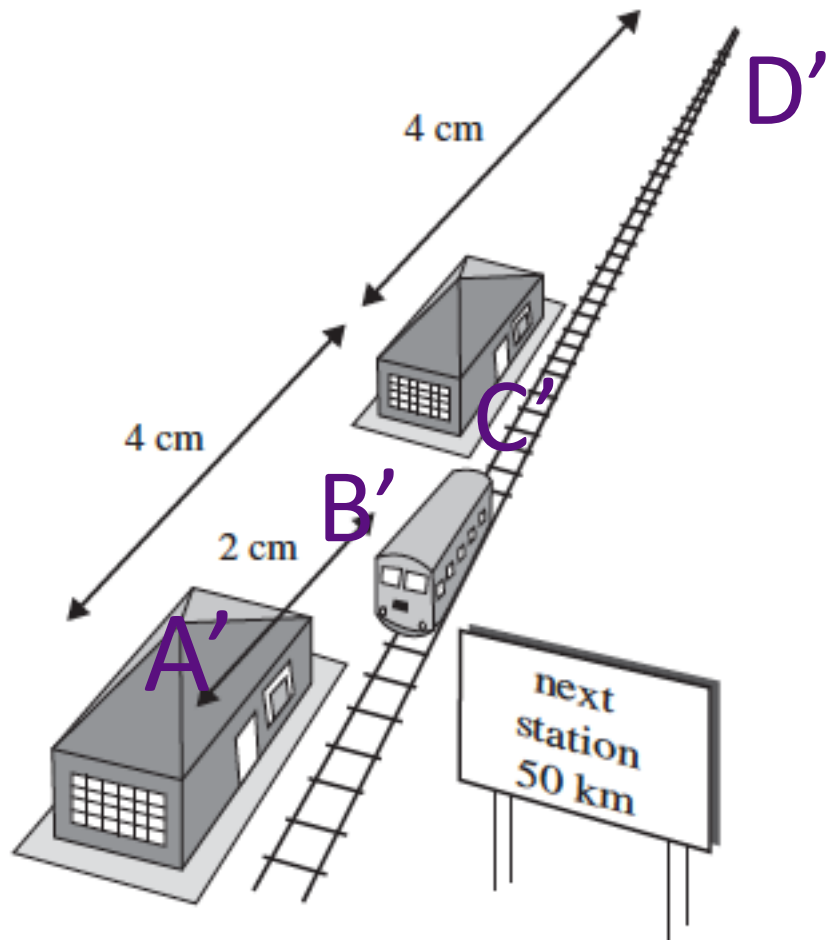


In pixels $\frac{A'C'}{A'D'} : \frac{B'C'}{B'D'} = ?$

When a point D is at infinity, the cross-ratio becomes a ratio !

$$\frac{AC}{AD} : \frac{BC}{BD} = \frac{AC}{BC} \quad \left(\text{Think } \frac{AC}{\infty} : \frac{BC}{\infty} = \frac{AC}{BC} \times \frac{\infty}{\infty} = \frac{AC}{BC} \right)$$

Vanishing points allow us to measure length ratios!



How far away is the train from the next station?
i.e. Length of BC ?

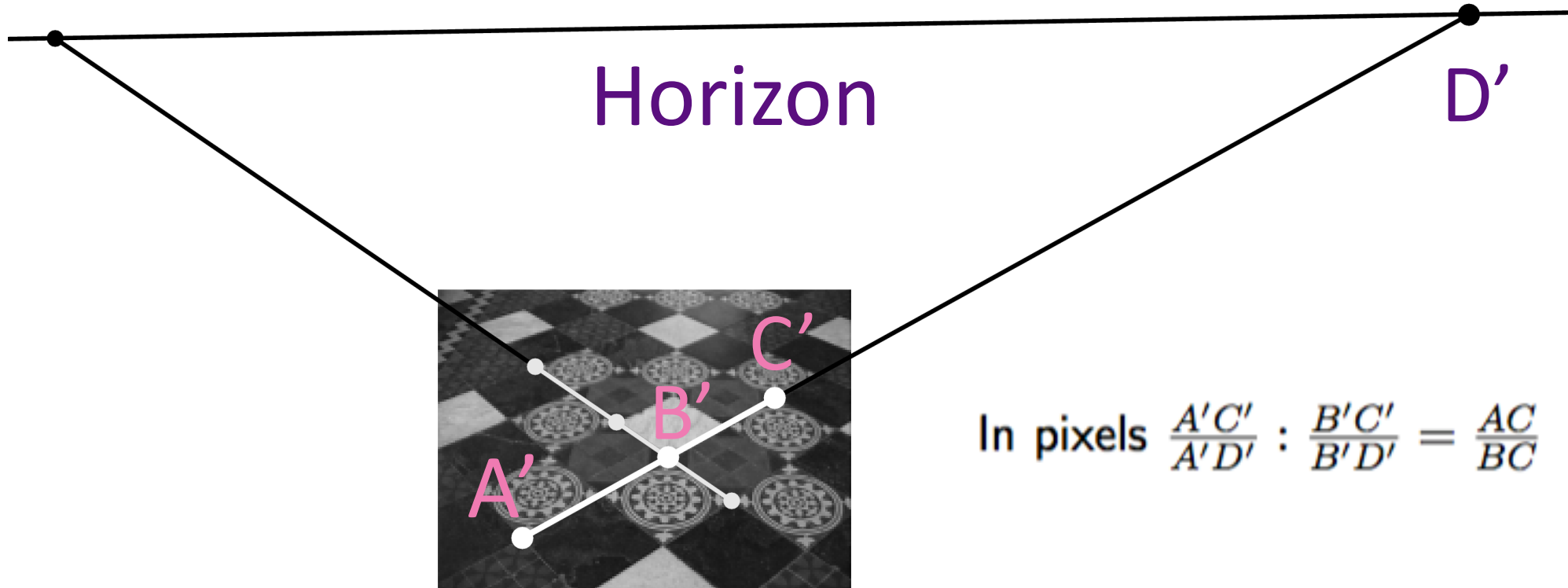
$$\frac{A'C'}{A'D'} : \frac{B'C'}{B'D'} = \frac{AC}{BC}$$

$$\frac{4}{8} : \frac{2}{6} = \frac{50}{BC}$$

$$\frac{4}{8} : \frac{2}{6} = \frac{50}{BC} \text{ and } BC = 33.33 \text{ km.}$$

Note: We only needed measurement of 1 distance in the real world.

Also, world plane length ratios determine vanishing points!



$$\frac{A'C'}{A'D'} : \frac{B'C'}{B'D'} = \frac{AC}{BC} = 2$$

Given: $AB=BC$

If we know A' , B' , C' in pixels, we can find D'

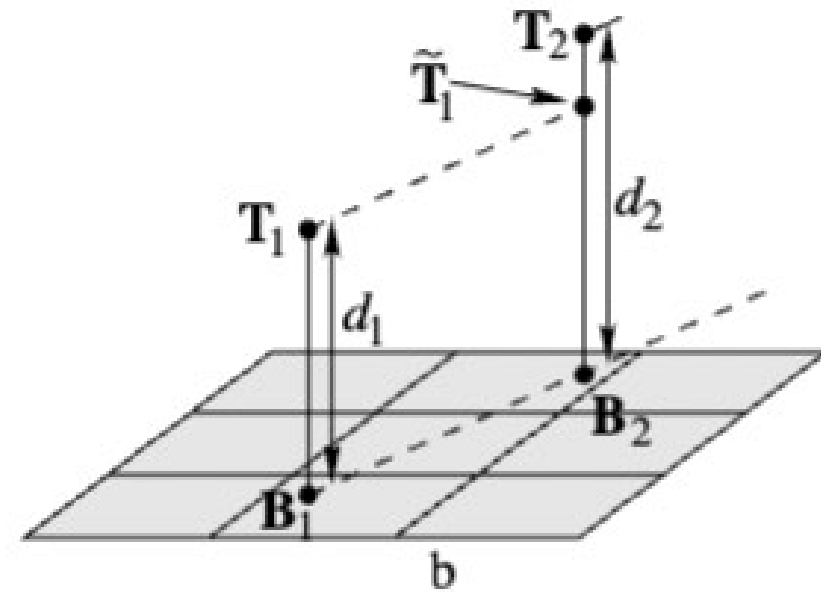
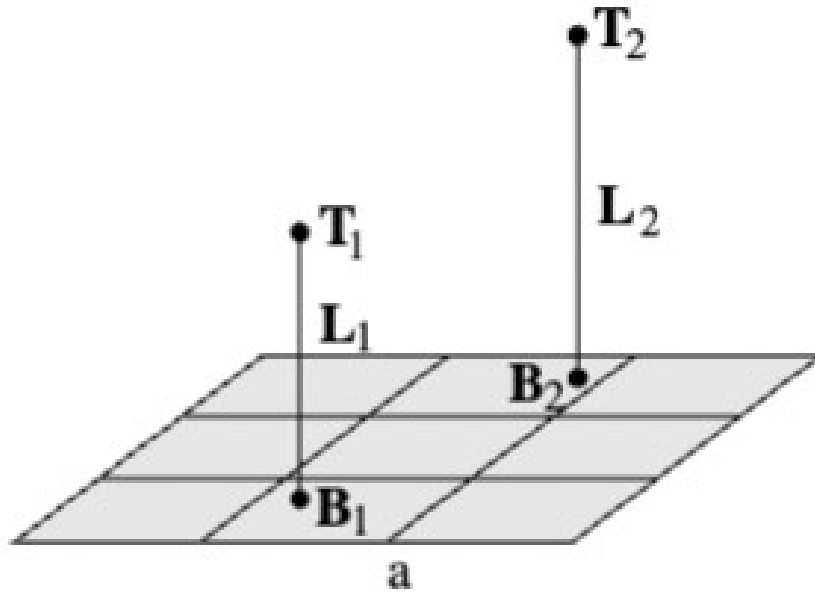
In this way, we can find vanishing points and the horizon without even needing parallel lines!

Measuring heights, i.e. distances *from* a world plane

- So far, we've been looking at distances *on* a world plane.
- Next, distances *off it*?

Length transfer in 3D

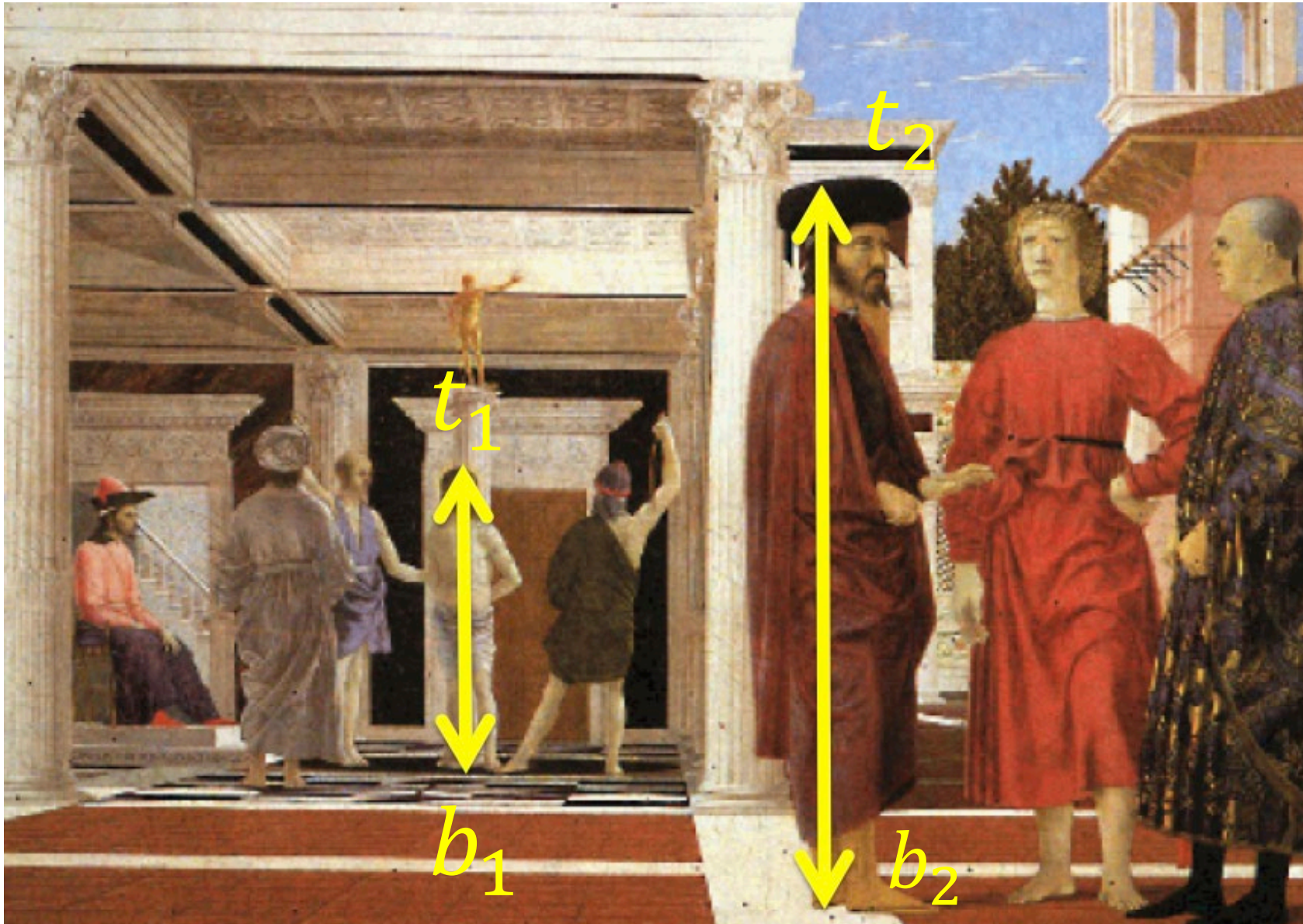
- In the real 3-D world, you can compare one object with known length to another to “transfer” its length.



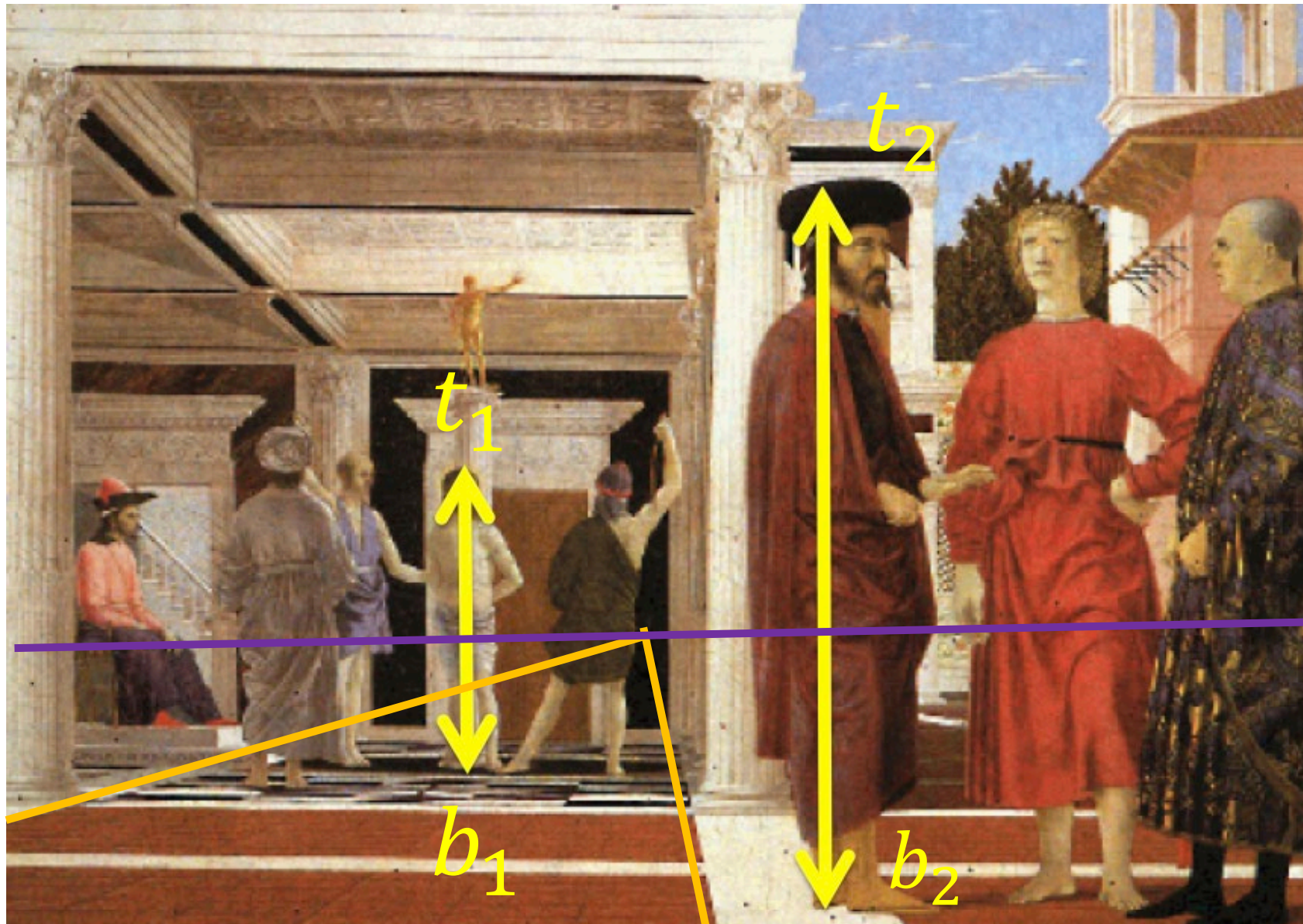
How to do this in an image?

Distance Transfer: How tall is the man if the statue is 180cm (in 3D world space)?

Note: we will be going back and forth between world points (denoted as capital letters) and their projections (denoted as small letters).

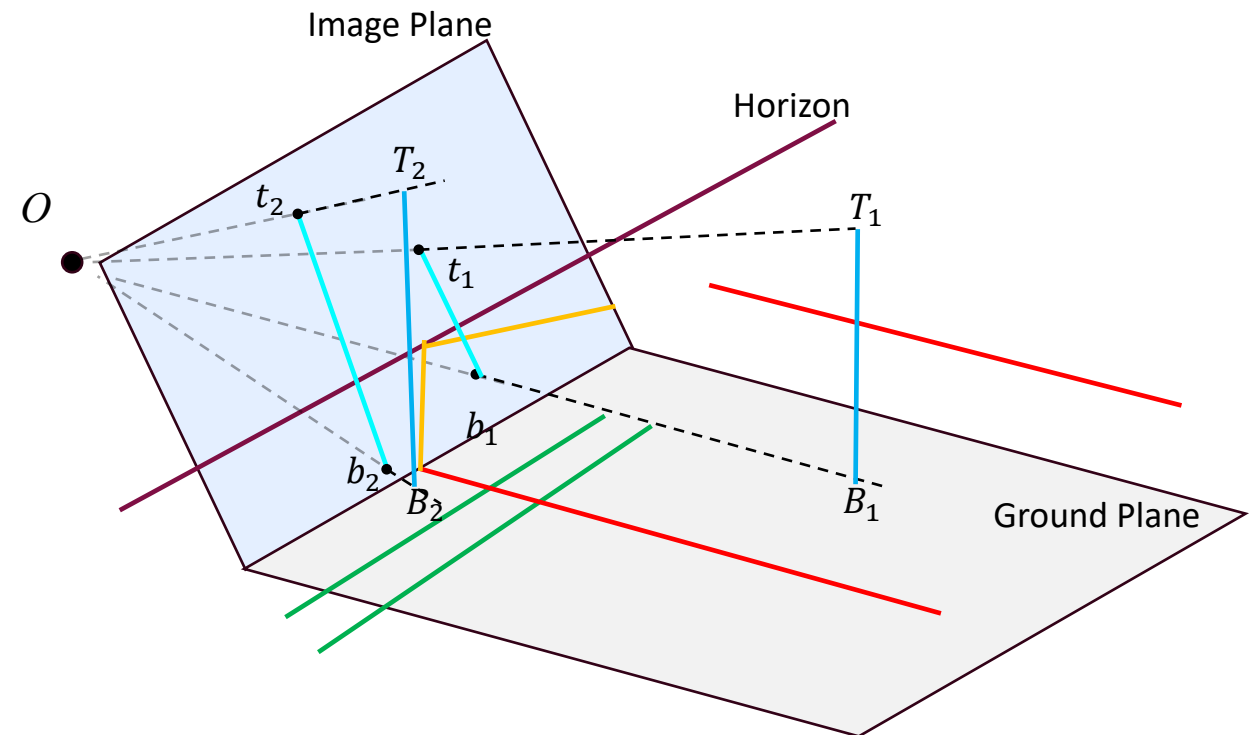
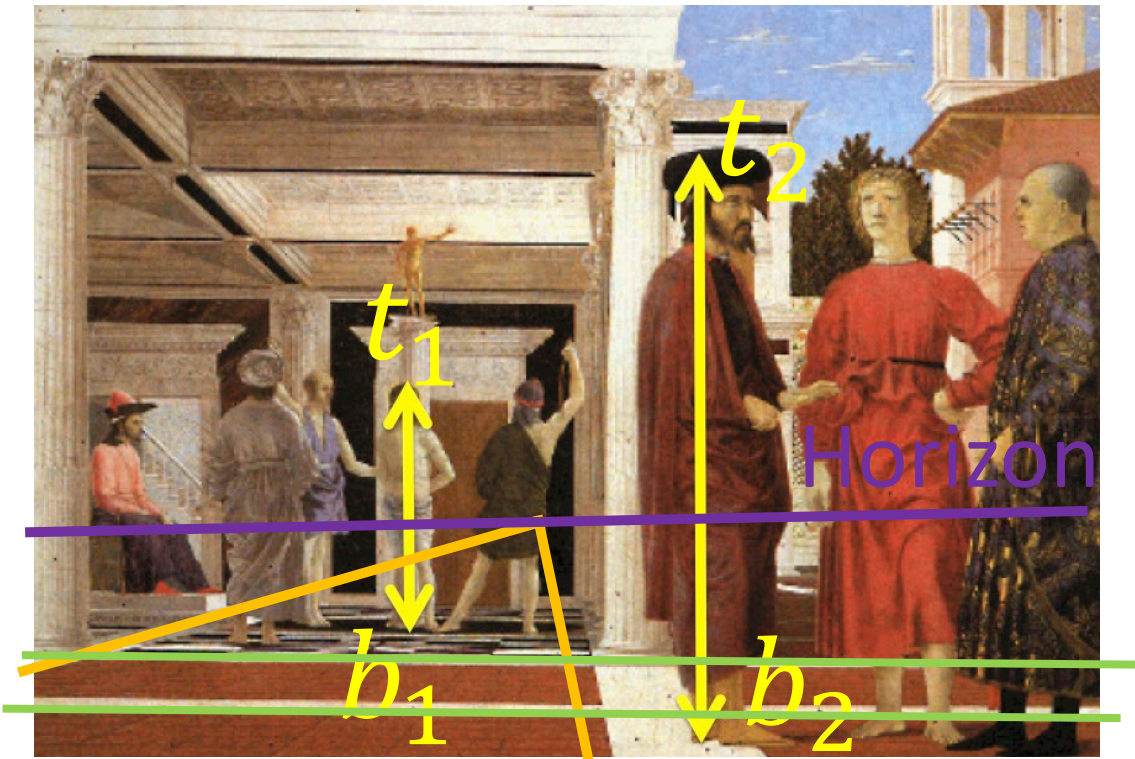


1. Find horizon (from homography, or by finding VPs)

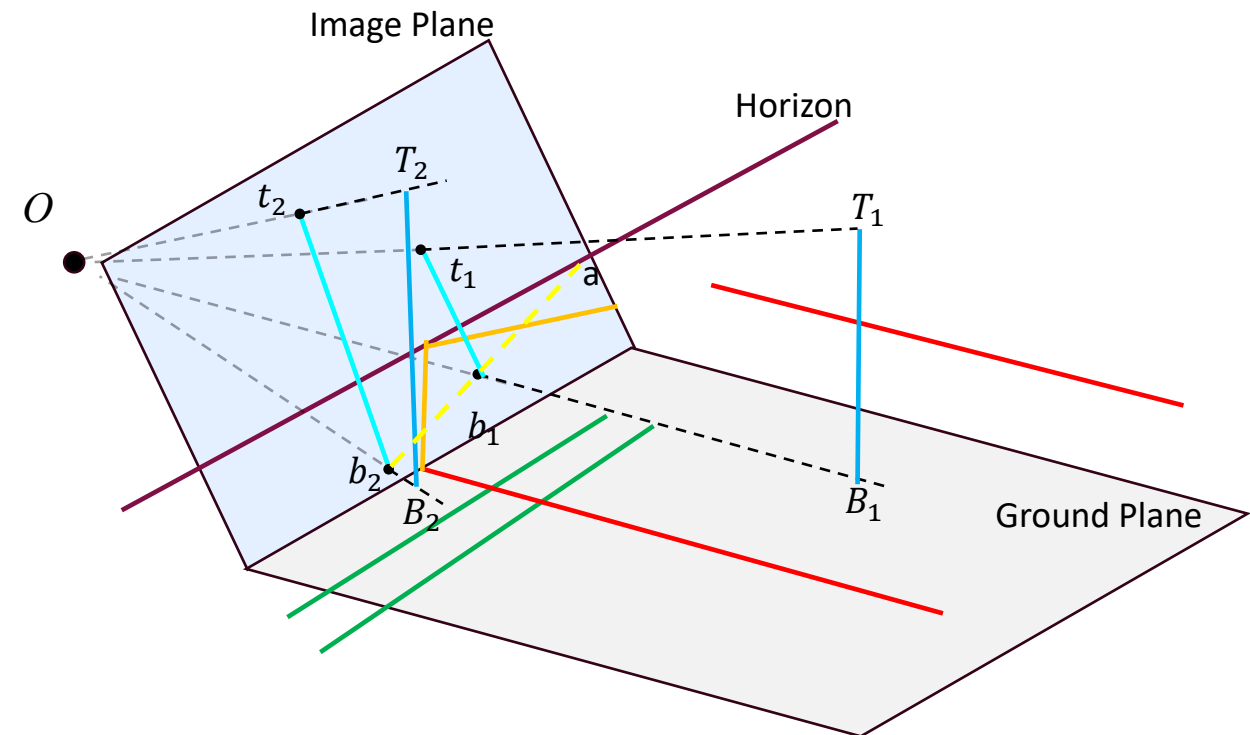
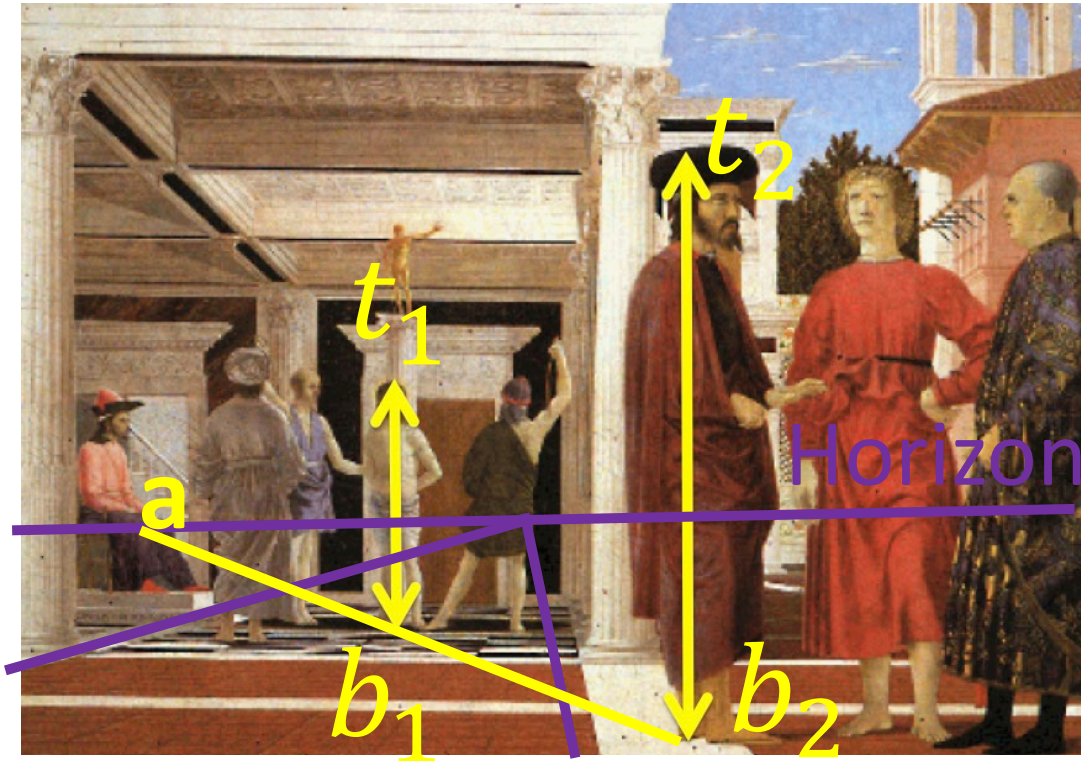


Horizon

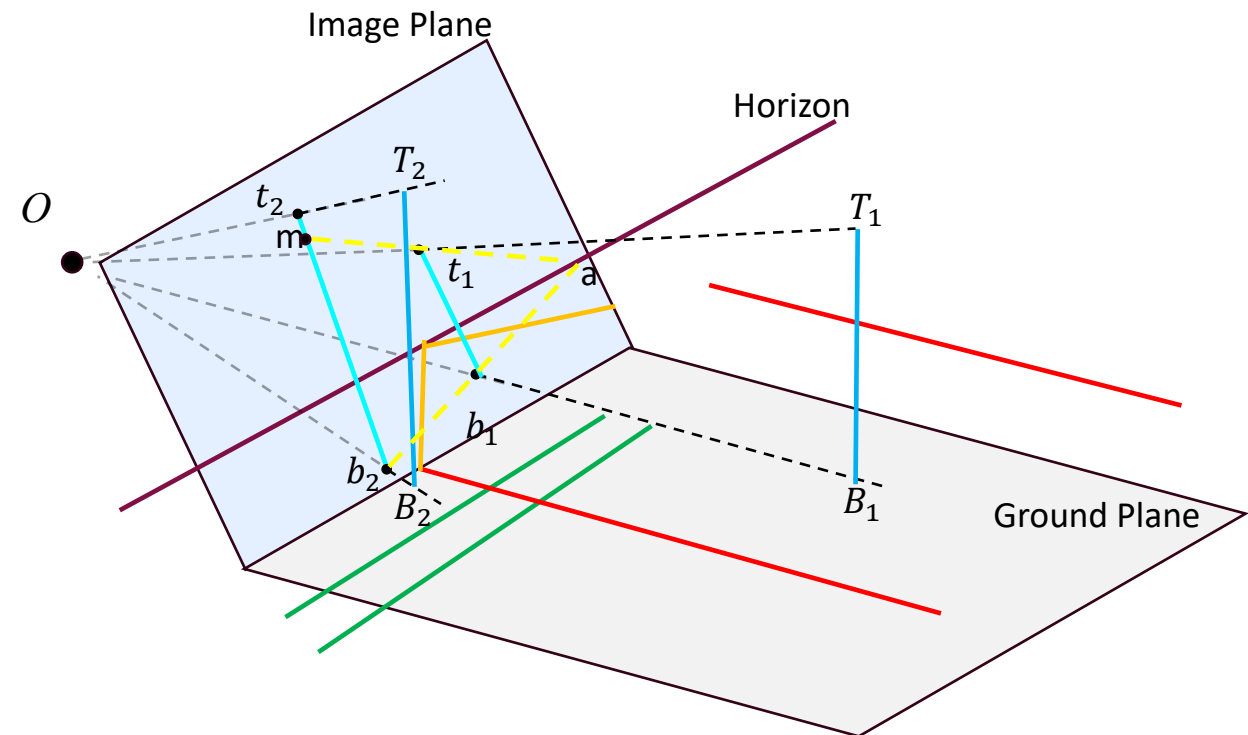
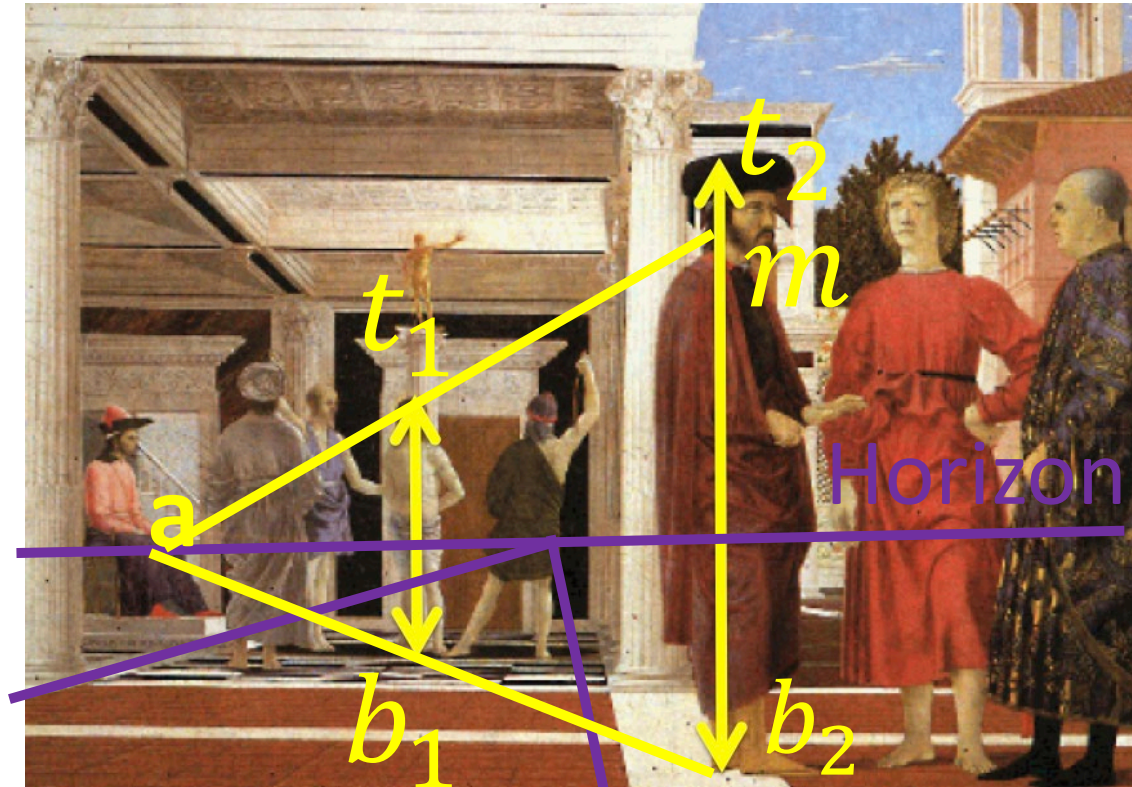
1. Find horizon (from homography, or by finding VPs)



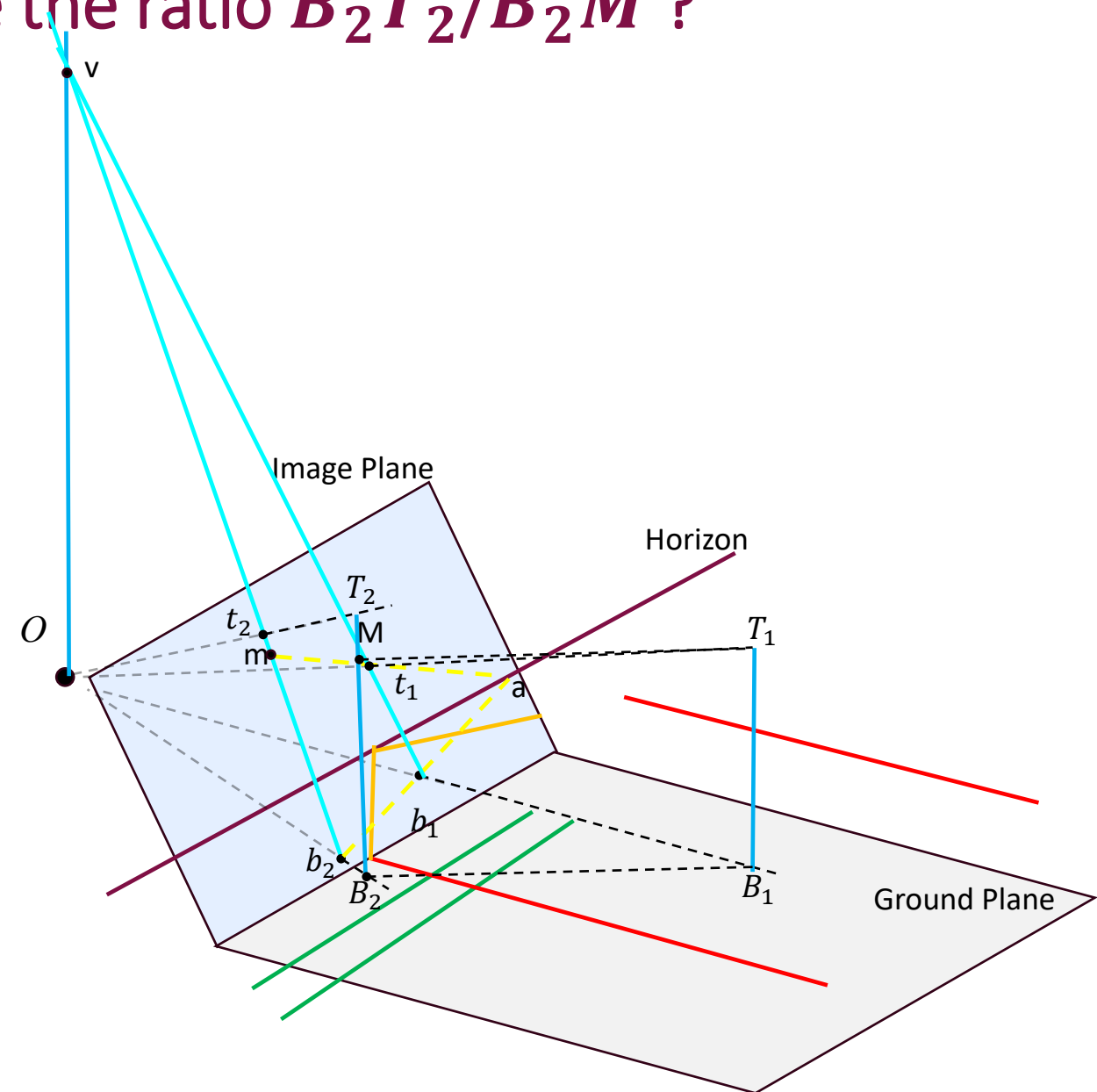
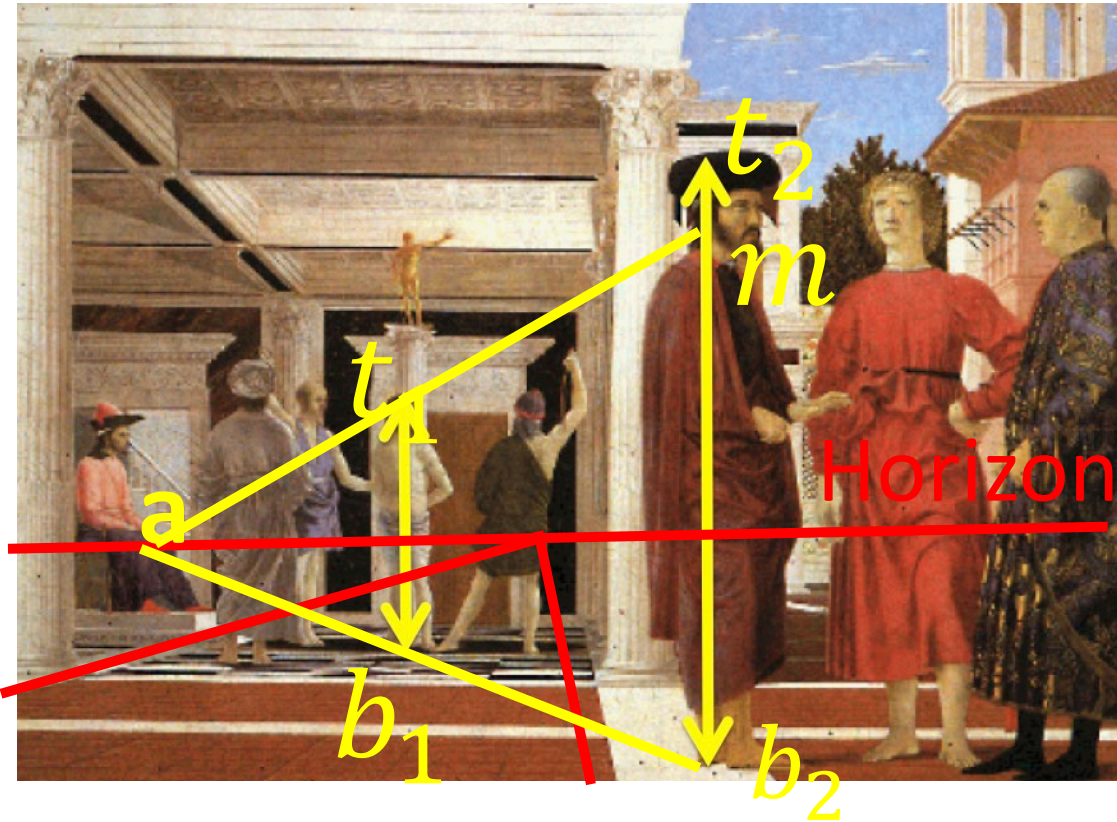
2. Connect the feet of the man and the statue and find intersection a' with horizon!



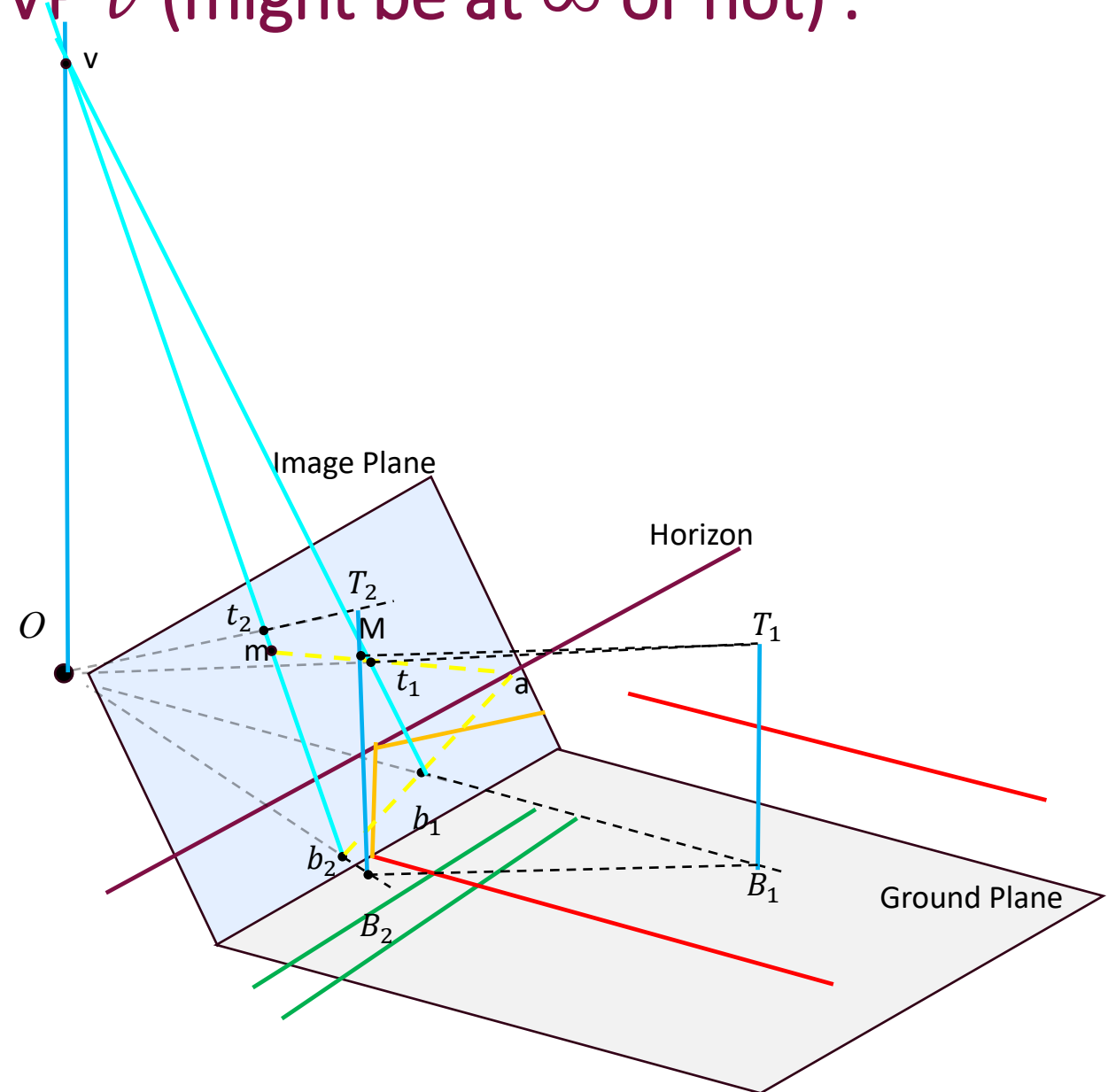
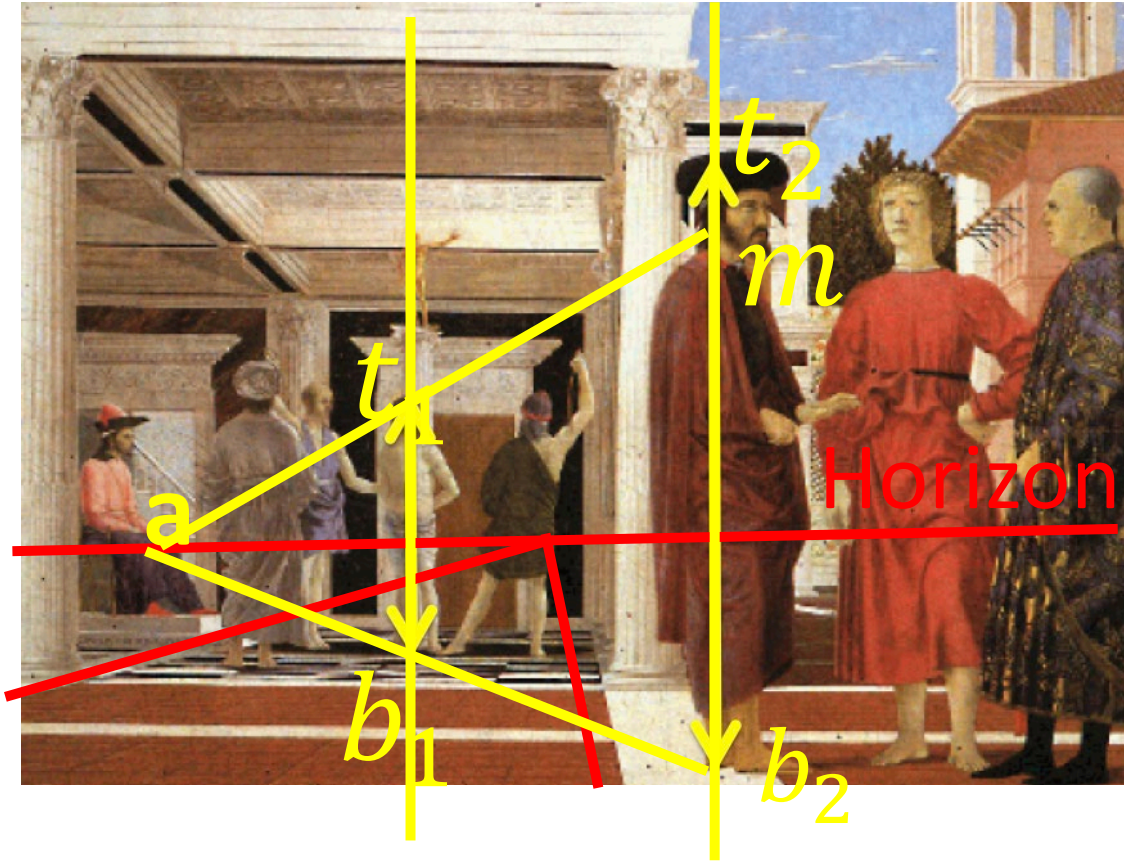
3. Connect a with top of statue t_1



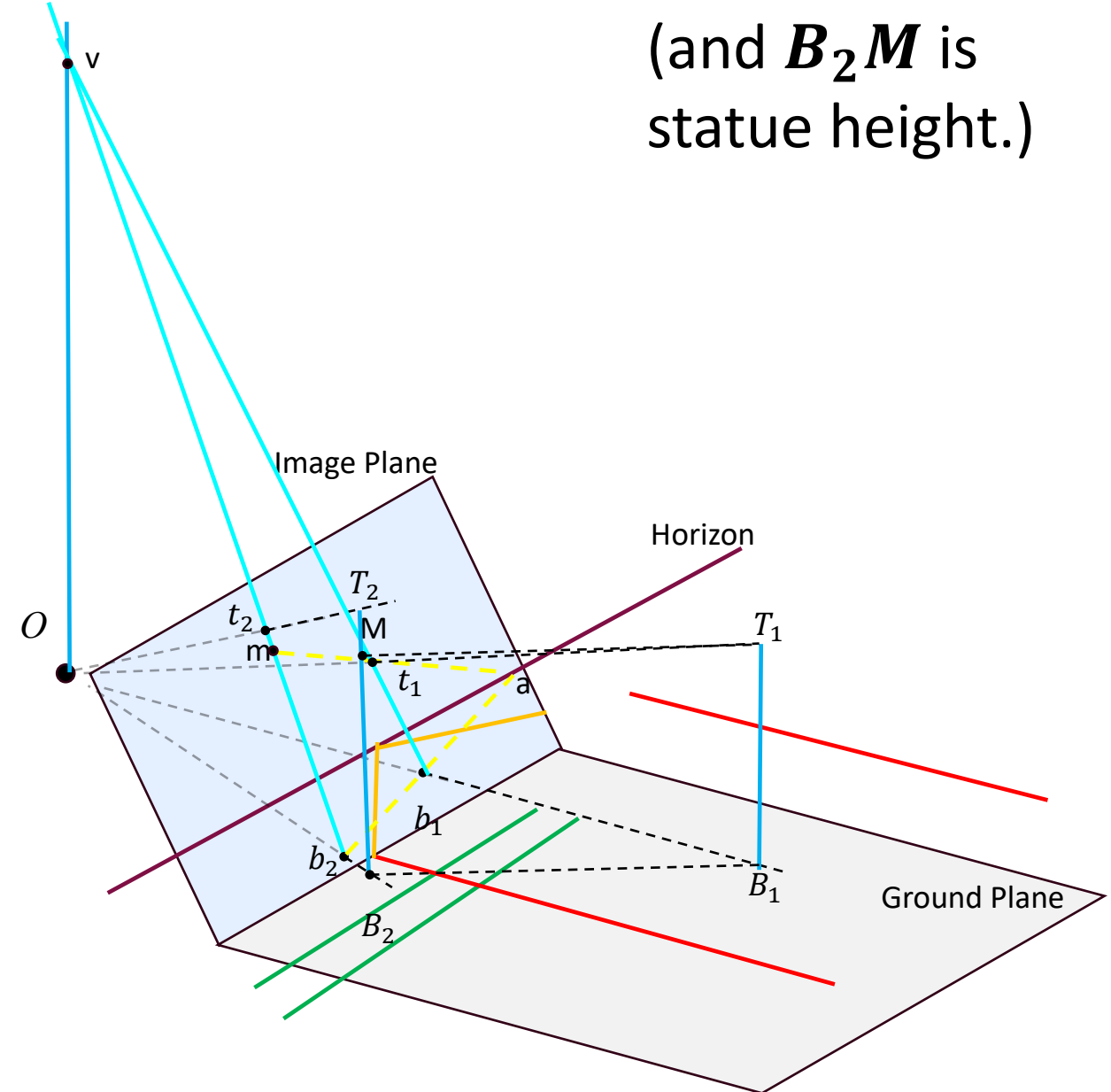
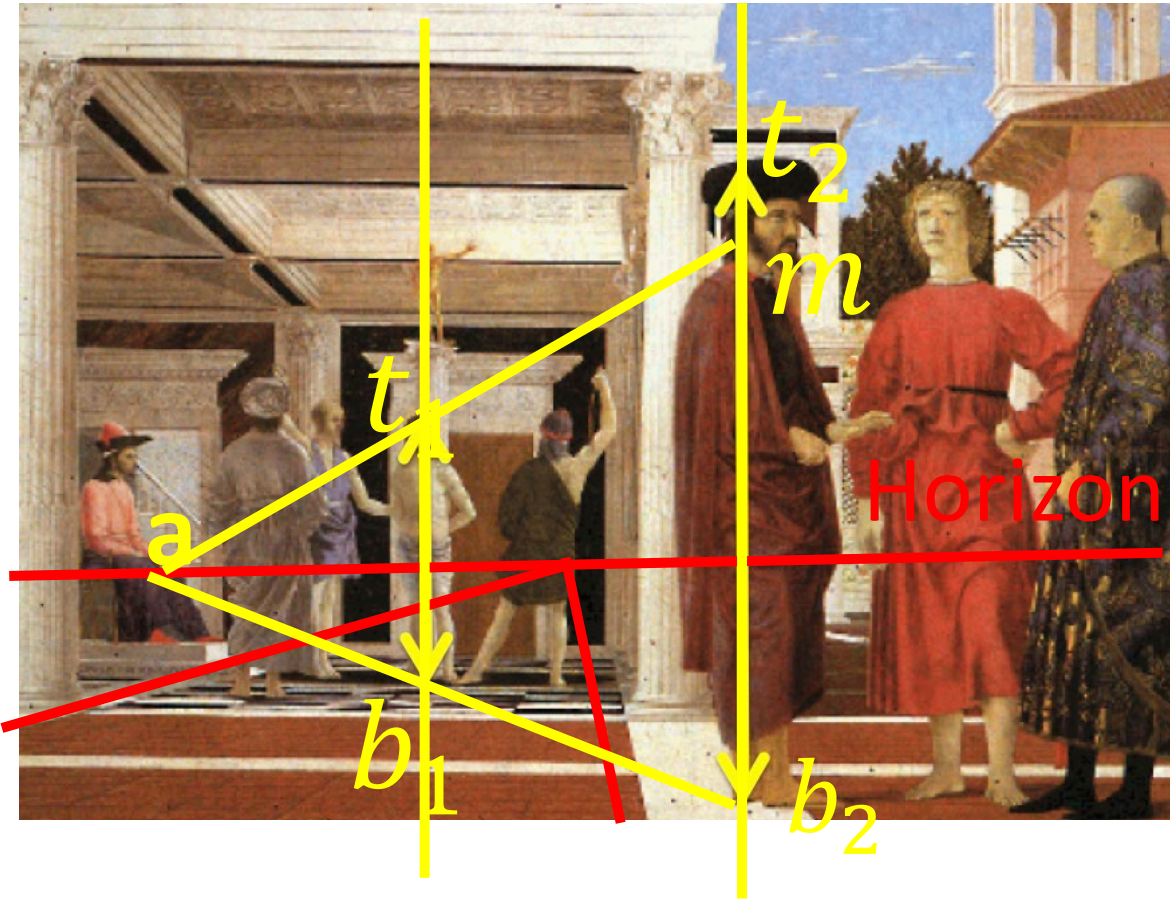
4. But we want B_2T_2 in the world.
How can we compute the ratio B_2T_2/B_2M ?



4. Only if we know a vanishing point in the vertical direction. Let b_1t_1 and b_2m intersect at a vertical VP v (might be at ∞ or not) .



5. Then cross ratio of $\{v, t_2, m, b_2\}$ = ratio $B_2T_2 : B_2M$
 (and B_2M is statue height.)



Single View Metrology via Cross Ratios

- If we know the vanishing point for a direction, we can compute any ratio along this direction!
- We can transfer lengths among parallel line segments in the world using knowledge of the vanishing point for their direction.
- All of this without explicitly computing any focal length, intrinsics, homographies etc.!

We can do “image forensics” on paintings or old photos!

See also: ZH Sec 8.7,
ZH example 8.25

How to Detect Faked Photos

Techniques that analyze the consistency of elements within an image can help to determine whether it is real or manipulated.

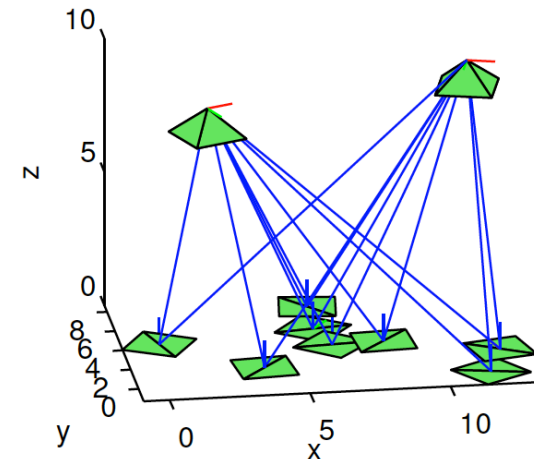
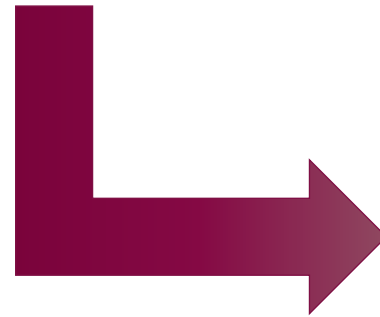
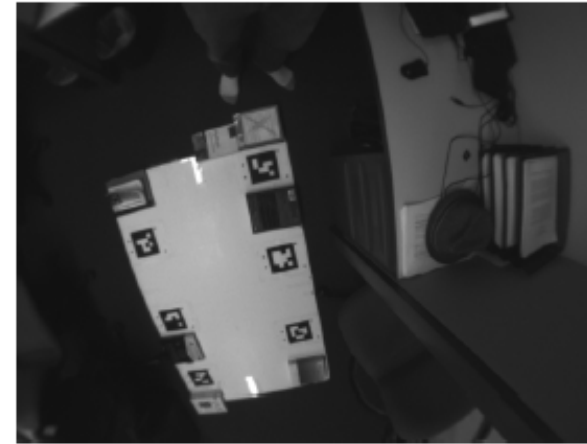
Fixing Camera 6DoF Pose Estimation w.r.t world plane

(given homography H & camera intrinsics K)

Camera 6DOF pose

- A camera's rotation (3DOF) and translation (3DOF) jointly is called its 6-DOF pose.
- “camera pose” estimation = finding the “extrinsics matrix”

Recall: we know how to find homography w.r.t. a planar pattern in the world.



Recall: homography gives pose (given intrinsics K)

Pose from Projective Transformation

Recall the projection from world to camera

$$\begin{pmatrix} u \\ v \\ w \end{pmatrix} \sim K \begin{pmatrix} r_1 & r_2 & r_3 & T \end{pmatrix} \begin{pmatrix} X \\ Y \\ Z \\ W \end{pmatrix}$$

and assume that all points in the world lie in the ground plane $Z = 0$.

Then the transformation reads

$$\begin{pmatrix} u \\ v \\ w \end{pmatrix} \sim K \begin{pmatrix} r_1 & r_2 & T \end{pmatrix} \begin{pmatrix} X \\ Y \\ W \end{pmatrix}$$

$$\text{And } r_3 = r_1 \times r_2$$

But actually, not quite!

- According to the previous slide $K(r_1 \ r_2 \ T) = H$, or in other words,

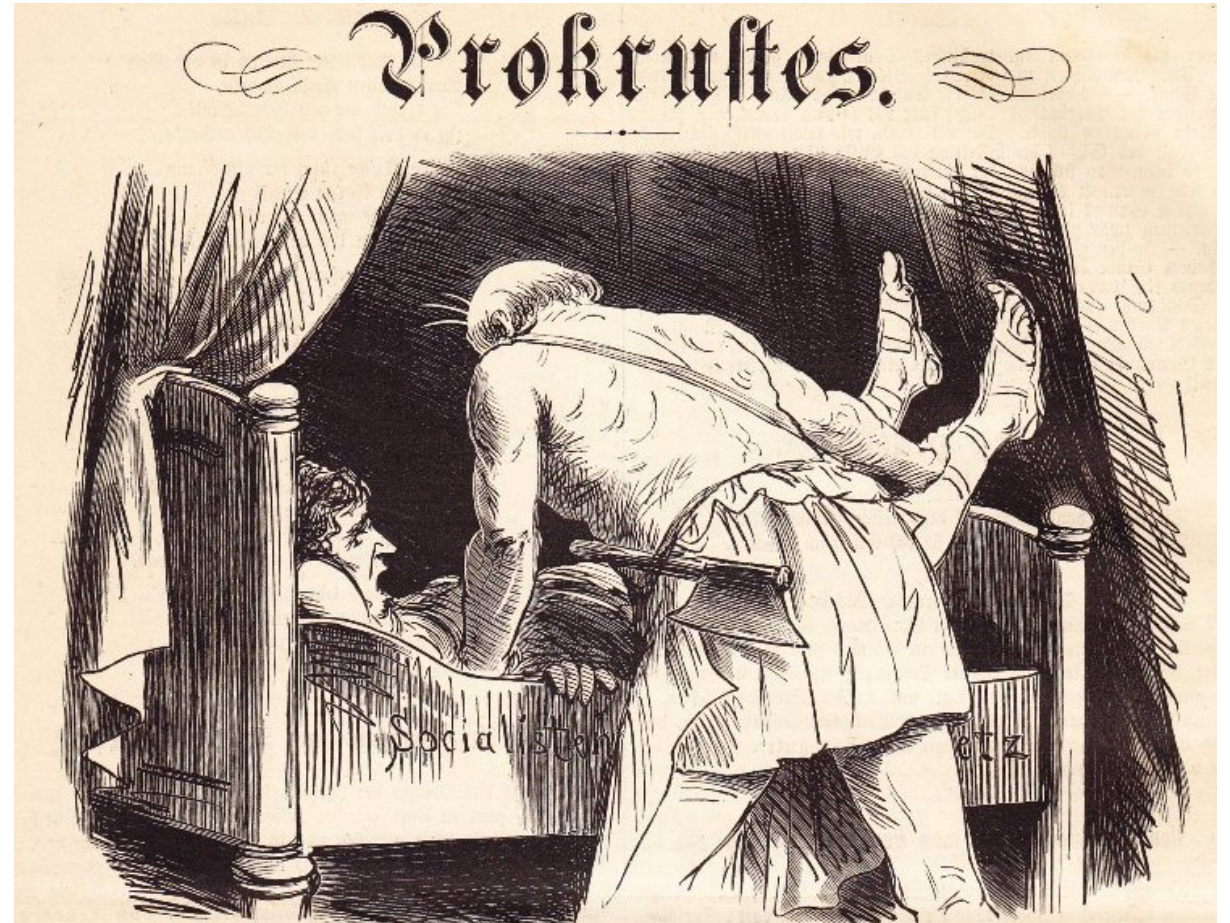
$$K^{-1}H = (r_1, r_2, T) \text{ and } r_3 = r_1 \times r_2$$

If only life were so simple!

- Problem: when we ***estimate*** homographies (e.g. through solving linear systems with $2n$ equations from $n \geq 4$ point correspondences), and then compute $K^{-1}H$, we aren't guaranteed to find a *valid* r_1 and r_2 pair. i.e. an orthonormal pair.
 - So, we need to find a way to first “correct” $(K^{-1}H)_{3 \times 3}$ to get orthonormal r_1 and r_2 . Often called the “Procrustes”, or “special orthogonal (SO) Procrustes” problem.
 - And we must solve this in real-time for robotics applications, so preferably an inexpensive approach.

The macabre Greek legend of Procrustes

We are trying to get every $(K^{-1}H)$ “traveler” to fit the “bed” of valid rotation matrices by stretching it or chopping it off.



Let us name the columns of $K^{-1}H$:

$$K^{-1}H = (h'_1 \quad h'_2 \quad h'_3)$$

We seek orthogonal r_1 and r_2 that are the closest to h'_1 and h'_2 . The solution to this problem is given by the Singular Value Decomposition.

We find the orthogonal matrix R that is the closest to $(h'_1 \quad h'_2 \quad h'_1 \times h'_2)$:

$$\arg \min_{R \in SO(3)} \|R - (h'_1 \quad h'_2 \quad h'_1 \times h'_2)\|_F^2$$

Kabsch algorithm for Procrustes

$$\arg \min_{R \in SO(3)} \|R - \begin{pmatrix} h'_1 & h'_2 & h'_1 \times h'_2 \end{pmatrix}\|_F^2$$

If the SVD of

$$\begin{pmatrix} h'_1 & h'_2 & h'_1 \times h'_2 \end{pmatrix} = USV^T$$

then the solution is

$$R = U \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & \det(UV^T) \end{pmatrix} V^T$$

The diagonal matrix is inserted to guarantee that $\det(R) = 1$.

To find the translation : $T = h'_3 / \|h'_1\|$

(In case original columns were not even unit norm)

proof in supp readings- Kabsch-Algorithm-RT-from-H-proof.pdf. We will also prove it in the next class.

Full Kabsch algorithm for finding pose via homography

1. Find H up to a scale factor from the point correspondences

2. Compute $H' = K^{-1}H$. Let H' 's columns be $(a \ b \ c)$

3. Minimize

$$\|(a \ b \ c) - \lambda(r_1 \ r_2 \ T)\|_F$$

w.r.t. $\lambda \in \mathbb{R}, r_1, r_2, T \in \mathbb{R}^3$

s.t. $r_1^T r_2 = 0$ and $\|r_1\| = \|r_2\| = 1$

Let

$$(a \ b) = U_{3 \times 2} \begin{pmatrix} s_1 & 0 \\ 0 & s_2 \end{pmatrix} V_{2 \times 2}^T.$$

Then

$$(r_1 \ r_2) = U_{3 \times 2} V_{2 \times 2}^T \quad \text{and} \quad \lambda = \frac{s_1 + s_2}{2}$$

Alternative to running Kabsch including the 3rd column $c = h'_1 \times h'_2$ as on last slide

4. $T = c/\lambda$ and $R = (r_1 \ r_2 \ r_1 \times r_2)$. Scale R to have determinant 1 if needed.

So now, camera pose (actually) known w.r.t world plane!

